

Interpret Division of Fractions

Flagship

Lesson 2-1-flagship

Name: _____

Date: _____

Class: _____

EVIDENCE LAB MISSION

The Fraction Vault

You are the lead forensic analyst at the Reyes Detective Agency. A 3-foot strip of evidence tape must be cut into exact $\frac{1}{4}$ -foot sections before it can be logged — and the vault that holds the case file only opens once you know how many pieces that makes. Interpret fraction division and crack the case.

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

In $3 \div \frac{1}{4} = 12$, the dividend is 3 — it is the total being split

Dividend

The number you are splitting up in a division problem.

In $3 \div \frac{1}{4} = 12$, the divisor is $\frac{1}{4}$ — it is the size of each piece

Divisor

The number you split by in a division problem.

In $3 \div \frac{1}{4} = 12$, the quotient is 12 — there are 12 quarter-size pieces in 3

Quotient

The answer when you divide.

The reciprocal of $\frac{1}{4}$ is $\frac{4}{1} = 4$. Multiplying by the reciprocal gives the same result as dividing.

Reciprocal

A fraction turned upside down.

$\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$ — each represents one equal part of a whole

Unit fraction

A fraction with 1 on top, like $\frac{1}{4}$.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can use models to interpret what it means to divide by a fraction.

1 Notice

Agent Reyes finds a 3-foot strip of tape at the crime scene.

2 Key idea

I can use models to interpret what it means to divide by a fraction.

3 Words I'll use

Dividend

Divisor

Quotient

Reciprocal

Unit fraction



Think About It

- What total length is being divided?
- What is the size of each section?
- Will the answer be greater than or less than 3?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: What does $3 \div \frac{1}{4}$ mean?

- A. How many $\frac{1}{4}$ -size pieces fit into 3
- B. 3 groups of $\frac{1}{4}$
- C. 3 minus $\frac{1}{4}$
- D. $\frac{1}{4}$ of 3

Step 1 $3 \div \frac{1}{4}$ asks: how many $\frac{1}{4}$ -size pieces fit into 3?

Step 2 The answer is 12, because each whole has 4 fourths, and $3 \times 4 = 12$.



Answer: A. How many $\frac{1}{4}$ -size pieces fit into 3

 We do — together

Solve this one with your class using the same steps.

Problem: Which expression means 'how many $\frac{1}{3}$ -size pieces are in 2'?

A. $2 \div \frac{1}{3}$

B. $2 \times \frac{1}{3}$

C. $\frac{1}{3} \div 2$

D. $2 + \frac{1}{3}$

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: A rope is 4 feet long. How many $\frac{1}{2}$ -foot pieces can be cut from it?

A. 8 pieces

B. 2 pieces

C. 4 pieces

D. $\frac{1}{2}$ piece

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

A 3-foot strip of evidence tape must be cut into $\frac{1}{4}$ -foot sections. Before calculating, will the number of pieces be more or fewer than 3? Explain your thinking.

Level 1 support

Start here: Because each piece is smaller than a whole foot, 3 feet makes more than 3 pieces.

 **Need a hint?**

Sentence starters (optional)

There will be ____ than 3 pieces because each piece is ____ than 1 foot.

Habr  ____ de 3 pedazos porque cada pedazo es ____ de 1 pie.

Dividing 3 by $\frac{1}{4}$ means asking how many ____ fit into ____.

Dividir 3 entre $\frac{1}{4}$ significa preguntar cu ntos ____ caben en ____.

Word bank: dividend divisor quotient unit fraction section

Listen for: Students reason that since $\frac{1}{4}$ is less than 1, more than 3 pieces fit, so the quotient is greater than the dividend.

Level 2

Why does dividing by a unit fraction like $\frac{1}{4}$ give an answer larger than the number you started with? Justify the idea, not just this case.

- Dividing by a number less than 1 makes the answer ____ because ____.
- This always happens when the divisor is ____.

EXPLORE

On your fraction bar, how many $\frac{1}{4}$ -foot sections fit into one whole foot? Point to where you see the pieces.

Level 1 support

Start here: Four $\frac{1}{4}$ -foot sections tile one whole foot, so each foot holds 4 pieces.



Need a hint?

Sentence starters (optional)

In one whole foot I see ____ sections of $\frac{1}{4}$ each.

En un pie completo veo ____ secciones de $\frac{1}{4}$ cada una.

So 3 feet hold ____ pieces because 3 times ____ equals ____.

Entonces 3 pies tienen ____ pedazos porque 3 por ____ es ____.

Word bank: **dividend** **divisor** **quotient** **unit fraction** **model**

Listen for: Students see 4 pieces per foot and connect it to $3 \times 4 = 12$ sections, matching $3 \div \frac{1}{4} = 12$.

Level 2

Use the reciprocal to show 3 divided by $\frac{1}{4}$ the algorithm way. Why does multiplying by 4 give the same answer as the bar model?

- Using the reciprocal, 3 times ____ equals ____ because ____.
- Multiplying by the reciprocal works because ____.

CONNECT

Where else outside the lab would you divide a whole number by a unit fraction, and how would you set it up?

Level 1 support

Start here: Any time you cut a whole amount into equal fractional-size pieces, you divide by a fraction.

 **Need a hint?**

Sentence starters (optional)

I would divide a whole number by a unit fraction when ____.

Dividiría un número entero entre una fracción unitaria cuando ____.

I would set it up as ____ divided by ____.

Lo plantearía como ____ entre ____.

Word bank: dividend divisor quotient reciprocal unit fraction

Listen for: Students give a real example (ribbon cut into $1/3$ -foot pieces, juice into $1/2$ -cup servings) and name the dividend and divisor.

Level 2

Write your own crime-scene word problem that needs dividing a whole number by a unit fraction, and explain what the quotient means in your story.

- My problem is ____, and the quotient tells how many ____.
- I know it needs division by a fraction because ____.

Write About the Math

The Writing Revolution

I can explain a fraction division using the words dividend, divisor, quotient, reciprocal, and unit fraction.

1. Kernel Sentence subject + verb

Model: Dividend is the number you are splitting up in a division problem.

Dividendo es el número que estás repartiendo en una división.

Write a kernel sentence about dividend. Use a subject and a verb.

Escribe una oración base sobre dividendo. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Dividend matters in math

Dividendo importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Dividend matters in math because ____.

Dividendo importa en matemáticas porque ____.

but
pero

Dividend matters in math, but ____.

Dividendo importa en matemáticas, pero ____.

so
entonces

Dividend matters in math, so ____.

Dividendo importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about dividend.
Di un hecho verdadero sobre dividend.

Dividend ____.

Question *Pregunta*

Ask a question about dividend.
Haz una pregunta sobre dividend.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about dividend.
Muestra entusiasmo sobre dividend.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with dividend.
Dile a un compañero qué hacer con dividend.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

There will be ____ **than 3 pieces because each piece is** ____ **than 1 foot.**

Habrá ____ *de 3 pedazos porque cada pedazo es* ____ *de 1 pie.*

Dividing 3 by $\frac{1}{4}$ means asking how many ____ **fit into** ____.

Dividir 3 entre $\frac{1}{4}$ significa preguntar cuántos ____ *caben en* ____.

In one whole foot I see ____ **sections of $\frac{1}{4}$ each.**

En un pie completo veo ____ *secciones de $\frac{1}{4}$ cada una.*

Try It

Solve on your own. Check the answer key when you are done.

1. What is $6 \div 1/5$?

- A. 30
- B. $6/5$
- C. $5/6$
- D. 1

Show your work:

2. A shelf is 5 feet long. Books are each $1/4$ foot wide. How many books fit on the shelf? Which expression represents this?

- A. $5 \div 1/4 = 20$ books
- B. $5 \times 1/4 = 1 \frac{1}{4}$ books
- C. $1/4 \div 5 = 1/20$ book
- D. $5 + 1/4 = 5 \frac{1}{4}$ books

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Dividing 6 by $\frac{1}{2}$ gives 12, but dividing 6 by 2 gives 3. How can dividing by a smaller number ($\frac{1}{2}$) give a bigger answer than dividing by a larger number (2)? Explain using a real-world example.

Sentence starter: When I divide 6 by 2, I am _____. When I divide 6 by $\frac{1}{2}$, I am _____. The answer is bigger because _____.

Show your work:

Reflect — Exit Ticket

What does $5 \div \frac{1}{4}$ mean, and what is the quotient?

- A. How many $\frac{1}{4}$ -size pieces fit into 5; quotient is 20
- B. 5 groups of $\frac{1}{4}$; quotient is $\frac{5}{4}$
- C. $\frac{1}{4}$ of 5; quotient is $\frac{5}{4}$
- D. 5 minus $\frac{1}{4}$; quotient is $4\frac{3}{4}$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $2 \div 1/3$ — *Finding how many $1/3$ -size pieces fit into 2 is represented by $2 \div 1/3$. The answer is 6.*
2. **You Try (notes):** A. 8 pieces — $4 \div 1/2 = 4 \times 2 = 8$ pieces. *Each whole foot contains 2 halves, and $4 \times 2 = 8$.*
3. **Try It 1:** A. 30 — $6 \div 1/5$ means how many $1/5$ -size pieces fit into 6. *Each whole has 5 fifths, so $6 \times 5 = 30$.*
4. **Try It 2:** A. $5 \div 1/4 = 20$ books — *You are finding how many $1/4$ -foot pieces fit into 5 feet, which is $5 \div 1/4 = 5 \times 4 = 20$ books.*
5. **Exit Ticket:** A. How many $1/4$ -size pieces fit into 5; quotient is 20 — $5 \div 1/4$ means 'how many $1/4$ -size pieces fit into 5.' *Since each whole has 4 fourths, $5 \times 4 = 20$.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Dividend is the number you are splitting up in a division problem.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).